# Comparing distributions

Training an AI image generator needs a number for how far its images are from real ones. Which distances still work when the two don't overlap?

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You will learn

- What we should ask of a number that says how far apart two distributions are.
- Why most ways of comparing two distributions stop working when the two put their probability in different places.
- How large the error of a trained generator can be, and why an objective that can underestimate that error is risky.
- How quickly more data brings us closer to the true distribution, and in which comparisons it never does.
- What maximum likelihood actually minimizes, and what that makes a model do when it cannot fit all of the data.

Before you start

- The empirical distribution and the pushforward from Chapter 1
- Why a generator with fewer latent dimensions than pixels has no density, also from Chapter 1
- The learning problem and its sources of error and the manifold hypothesis, also from Chapter 1

Gradient descent can only improve a generator if we give it a number to make smaller. That number has to say how far the generator's images are from real ones, and this chapter is about how to choose it.

The number has to compare two whole collections. No single image can be right or wrong, since any digit is fine as long as the generator's digits taken together look like the real ones. In the notation of Chapter 1, the number compares the distribution $\mathbb{P}_g$ of the generator with the data distribution $\mathbb{P}^*$. On the circle of Chapter 1 these are two clouds of points in the plane. As the generator's cloud moves toward the circle the number should go down, and it should reach zero only when the generator produces exactly the distribution of the data.

## What we ask of a distance

We want to turn "how far $\mathbb{P}_g$ is from $\mathbb{P}^*$" into a number, and we already know one kind of number like that. The distance between two points of the plane is zero only when the points coincide, it is the same in both directions, and a detour through a third point never makes the

trip shorter. These three properties are what mathematicians keep when they make “distance” precise for other objects, under the name *metric*.

**Definition 1** (Metric). A *metric* on a set $\mathcal{P}$ is a function $\rho : \mathcal{P} \times \mathcal{P} \to [0, +\infty)$ such that for all $x, y, z \in \mathcal{P}$ we have $\rho(x, y) = 0$ if and only if $x = y$, $\rho(x, y) = \rho(y, x)$, and the triangle inequality $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

Here $\mathcal{P}$ is a set of probability distributions on $\mathbb{R}^d$. The first axiom is the one that makes the zero a correct target for training. The distance must see every difference between two distributions, including differences that a summary would miss. Judging two distributions by their means alone would violate it, since two very different distributions can share a mean and would then sit at distance zero. Many useful comparisons keep this axiom but give up symmetry or the triangle inequality, so we also need a weaker notion.

**Definition 2** (Divergence). A *divergence* is a function $D(\mathbb{P} \,\|\, \mathbb{Q}) \in [0, +\infty]$ such that $D(\mathbb{P} \,\|\, \mathbb{Q}) = 0$ if and only if $\mathbb{P} = \mathbb{Q}$.

A divergence may be asymmetric, may break the triangle inequality and may be infinite. It still defines a target for training, since it is never negative and its only zero is at $\mathbb{P} = \mathbb{Q}$. Every metric is a divergence.

## Four ways to compare

We start with the comparison that needs the least machinery. Pick an event $A$, such as a region of the plane, and ask how much probability each distribution gives it. If some event gets very different probabilities, the two distributions are easy to tell apart, so we take the largest disagreement over all events.

To write this as an integral, and to write the comparisons after it, we need one piece of notation. The empirical distribution and a generator with $k < d$ have no density on $\mathbb{R}^d$ in the sense of Chapter 1, so we need something that plays the role of a density for them too. We measure each distribution against a common reference measure $\nu$ and call the result its density with respect to $\nu$, written $p_{\mathbb{P}}$ and $p_{\mathbb{Q}}$. For two distributions with ordinary densities $\nu$ is volume, and the formulas below are ordinary integrals. For distributions on finitely many points, $\nu$ counts points and integrals become sums. For any two distributions, the choice $\nu = \mathbb{P} + \mathbb{Q}$ works.

Going further

**The reference measure**

None of the results below needs $\nu$ written out, but one case shows how it works. Take the empirical distribution $\mathbb{P}_n$ and a distribution $\mathbb{P}^*$ with an ordinary density, and $\nu = \mathbb{P}_n + \mathbb{P}^*$. Then $\mathbb{P}_n$ has the density $1$ on the training points and $0$ elsewhere, and $\mathbb{P}^*$ has the density $0$ on the

training points and 1 elsewhere.

The largest disagreement on a single event is *total variation*,

$$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = \sup_A |\mathbb{P}(A) - \mathbb{Q}(A)| = 1 - \int \min(p_{\mathbb{P}}, p_{\mathbb{Q}})\, d\nu.$$

The first form is the definition we just gave. The supremum is reached at $A = \{p_{\mathbb{P}} > p_{\mathbb{Q}}\}$, the event that $\mathbb{P}$ favours most. Its disagreement is the total excess of $p_{\mathbb{P}}$ over $p_{\mathbb{Q}}$, which is all the mass of $\mathbb{P}$ minus the part it shares with $\mathbb{Q}$, and that gives the second form. The integral there is the mass the two distributions share, and TV is whatever is left after removing it. Two equal distributions share all their mass, so TV is $0$. Two distributions that share no mass have TV equal to $1$. TV is a metric with values in $[0, 1]$, and it depends only on the mass that the two distributions share, a fact that will cost us in the next section.

TV gives one number for one worst event. A finer comparison asks at each point how many times more likely it is under one distribution than under the other. The *Kullback-Leibler divergence* averages the logarithm of that ratio under $\mathbb{P}$,

$$\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = \int p_{\mathbb{P}} \log \frac{p_{\mathbb{P}}}{p_{\mathbb{Q}}}\, d\nu = \mathbb{E}_{X \sim \mathbb{P}} \left[ \log \frac{p_{\mathbb{P}}(X)}{p_{\mathbb{Q}}(X)} \right].$$

Two coins show how it behaves. Let $\mathbb{P}$ give heads probability $1/2$ and $\mathbb{Q}$ give heads probability $3/4$. Then $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = \frac{1}{2} \log \frac{2}{3} + \frac{1}{2} \log 2 \approx 0.14$, while $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) = \frac{3}{4} \log \frac{3}{2} + \frac{1}{4} \log \frac{1}{2} \approx 0.13$, so the two directions differ. If $\mathbb{Q}$ were a coin that never lands tails, then $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = +\infty$ because the ratio at tails is infinite, while $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) = \log 2$. In general KL is not symmetric, and it equals $+\infty$ when $\mathbb{P}$ puts mass on a set where $\mathbb{Q}$ puts none. Its unit is the *nat* because we use natural logarithms throughout.

Going further

**KL without a reference measure**

The usual way to write KL does not mention $\nu$. We write $\mathbb{P} \ll \mathbb{Q}$, and say that $\mathbb{P}$ is *absolutely continuous* with respect to $\mathbb{Q}$, when every event with $\mathbb{Q}(A) = 0$ also has $\mathbb{P}(A) = 0$. The Radon-Nikodym theorem then gives a function $d\mathbb{P}/d\mathbb{Q}$, the *Radon-Nikodym derivative*, such that $\mathbb{P}(A) = \int_A \frac{d\mathbb{P}}{d\mathbb{Q}}\, d\mathbb{Q}$ for every event $A$. It equals our ratio $p_{\mathbb{P}}/p_{\mathbb{Q}}$ except on a set of probability $0$ under both distributions, so

$$\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = \mathbb{E}_{X \sim \mathbb{P}} \left[ \log \frac{d\mathbb{P}}{d\mathbb{Q}}(X) \right].$$

If $\mathbb{P} \not\ll \mathbb{Q}$, this derivative does not exist and the divergence is $+\infty$. Our formula with $\nu$ gives the same answer, because the ratio is infinite on a set to which $\mathbb{P}$ gives positive probability. The coins show both cases. When $\mathbb{Q}$ never lands tails, $\mathbb{P} \not\ll \mathbb{Q}$, because $\mathbb{P}$ gives tails probability $1/2$. In the other direction $\mathbb{Q} \ll \mathbb{P}$, with $d\mathbb{Q}/d\mathbb{P}$ equal to $2$ at heads and $0$ at tails, and that gives the value $\log 2$.

Some of the terms in the sum can be negative, as in the coin example, but the total never is.

> **Proposition 3** (Gibbs' inequality). *For all distributions $\mathbb{P}$ and $\mathbb{Q}$, $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) \geq 0$ with equality if and only if $\mathbb{P} = \mathbb{Q}$. So KL is a divergence.*

*Proof.*

Jensen's inequality says that for a concave function such as the logarithm, $\mathbb{E}[\log Y] \leq \log \mathbb{E}[Y]$. Applied to $Y = p_{\mathbb{Q}}(X)/p_{\mathbb{P}}(X)$ with $X \sim \mathbb{P}$ it gives

$$-\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = \mathbb{E}_{X \sim \mathbb{P}}\left[\log \frac{p_{\mathbb{Q}}(X)}{p_{\mathbb{P}}(X)}\right] \leq \log \mathbb{E}_{X \sim \mathbb{P}}\left[\frac{p_{\mathbb{Q}}(X)}{p_{\mathbb{P}}(X)}\right] = \log \int_{\{p_{\mathbb{P}} > 0\}} p_{\mathbb{Q}}\, d\nu \leq \log 1 = 0.$$

Equality in Jensen's inequality for the strictly concave logarithm requires the ratio to be constant where $p_{\mathbb{P}} > 0$. The last inequality is an equality only if $\mathbb{Q}$ puts all its mass there. Together they give $\mathbb{P} = \mathbb{Q}$. □

KL has the two defects we just saw, since it can be infinite and it depends on the direction. A simple repair is to compare each distribution with their average $\mathbb{M} = \frac{1}{2}(\mathbb{P} + \mathbb{Q})$. The average gives mass wherever either of them does, so the ratio can no longer blow up. This is the *Jensen-Shannon divergence*,

$$\mathrm{JS}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2}\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{M}) + \frac{1}{2}\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{M}).$$

It is symmetric and always finite. For every set $A$ we have $\mathbb{P}(A) \leq 2\mathbb{M}(A)$, so the density ratio inside each KL is at most $2$ and $\mathrm{JS} \leq \log 2 \approx 0.693$.

All three comparisons so far look at how much probability each distribution gives to the same place. None of them asks how far the probability would have to move. The *Wasserstein distance* measures that, the cost of reshaping one distribution into the other. If each is a pile of sand of total mass $1$, it is the least average distance the sand must travel to turn one pile into the other, which is why it is also called the earth mover's distance.

To make this precise we need a way to describe a plan for moving the sand. A *coupling* of $\mathbb{P}$ and $\mathbb{Q}$ is a joint distribution of a pair $(X, Y)$ with $X \sim \mathbb{P}$ and $Y \sim \mathbb{Q}$, and we write $\Pi(\mathbb{P}, \mathbb{Q})$ for the set of couplings. A coupling is the transport plan made precise, because the joint probability of a pair $(x, y)$ says how much of the mass at $x$ is sent to $y$. Couplings always exist, since drawing $X$ and $Y$ independently gives one. Then

$$W_1(\mathbb{P}, \mathbb{Q}) = \inf_{\pi \in \Pi(\mathbb{P}, \mathbb{Q})} \mathbb{E}_{(X,Y)\sim\pi} \|X - Y\|$$

is the smallest average distance that mass has to travel. Two point masses $\delta_a$ and $\delta_b$ have only one coupling, which sends all the mass from $a$ to $b$, so $W_1(\delta_a, \delta_b) = \|a - b\|$. Unlike the other three, this keeps growing after the two distributions share no mass. Its definition also uses no density or reference measure. Squaring the distance in this formula and taking the square root of the result gives a second distance, $W_2$, which later chapters use.

Going further

**Fine print on JS and the Wasserstein distances**

JS is not a metric, although its square root is one (Endres and Schindelin 2003). The distance $W_2$ is the square root of the smallest average squared distance,

$$W_2(\mathbb{P}, \mathbb{Q}) = \Big( \inf_{\pi \in \Pi(\mathbb{P}, \mathbb{Q})} \mathbb{E}_{(X,Y)\sim\pi} \|X - Y\|^2 \Big)^{1/2},$$

and it appears later in FID (Chapter 6) and flow matching (Chapter 8). The distance $W_1$ is a metric on distributions with $\mathbb{E}\|X\| < \infty$ and $W_2$ on those with $\mathbb{E}\|X\|^2 < \infty$, so that the average distances are finite (Villani 2009). Neither needs a density, and both stay informative when a generator with $k < d$ misses the data.

The table collects what we have. Its last column gives the value when two distributions share no mass, and the next section shows that this is the situation of a generator with fewer latent dimensions than pixels.

| | Symmetric | Metric | Value when the two share no mass |
|---|---|---|---|
| TV | yes | yes | $1$, whatever the separation |
| KL | no | no | $+\infty$ |
| JS | yes | no | $\log 2$, whatever the separation |
| $W_1$ | yes | yes | grows with the separation |

## When two distributions share no mass

The last column of the table is a claim, and we can test it on an example small enough to compute. A divergence gives training a target, its zero, but gradient descent also has to be able to reach it. For that we want a number that keeps growing as two distributions move apart, because that growth is what gradient descent follows.

Take two uniform distributions of width $\Delta$ on the real line, one shifted by $\theta$ from the other. I chose uniforms because every quantity has a closed form for them and their supports separate at a definite shift. Here the *support* of a distribution is the smallest closed set that carries all its mass, which for a uniform is its interval. Two Gaussians overlap for every shift, and two point masses go from equal to separated with nothing in between.

> **Proposition 4** (Two shifted uniforms). *Let* $\mathbb{P} = \mathcal{U}[-\frac{\Delta}{2}, \frac{\Delta}{2}]$, $\mathbb{Q} = \mathcal{U}[\theta - \frac{\Delta}{2}, \theta + \frac{\Delta}{2}]$ *and* $r = \min(|\theta|/\Delta, 1)$. *Then*
>
> $$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = r, \qquad \mathrm{JS}(\mathbb{P}, \mathbb{Q}) = r \log 2, \qquad W_1(\mathbb{P}, \mathbb{Q}) = |\theta|,$$
>
> *and* $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = +\infty$ *for every* $\theta \neq 0$.

*Proof.*

Suppose $0 < |\theta| < \Delta$. Both densities equal $1/\Delta$ on an overlap of length $\Delta - |\theta|$, and each one is alone on an interval of length $|\theta|$. The shared mass is $(\Delta - |\theta|)/\Delta$, so $\mathrm{TV} = |\theta|/\Delta$. For JS, the overlap contributes nothing to $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{M})$ because $p_{\mathbb{P}} = p_{\mathbb{M}}$ there. Where $\mathbb{P}$ is alone the ratio is $2$, which contributes $\frac{|\theta|}{\Delta} \log 2$. The same holds for $\mathbb{Q}$, so $\mathrm{JS} = \frac{|\theta|}{\Delta} \log 2$. When $|\theta| \geq \Delta$ the distributions share no mass, so $\mathrm{TV} = 1$ and $\mathrm{JS} = \log 2$. For $W_1$, every coupling satisfies $\mathbb{E}|X - Y| \geq |\mathbb{E}X - \mathbb{E}Y| = |\theta|$, and moving every point by $\theta$ achieves this bound. Finally, for $\theta \neq 0$ the distribution $\mathbb{P}$ puts positive mass on an interval where $\mathbb{Q}$ puts none, so $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{Q}) = +\infty$. □

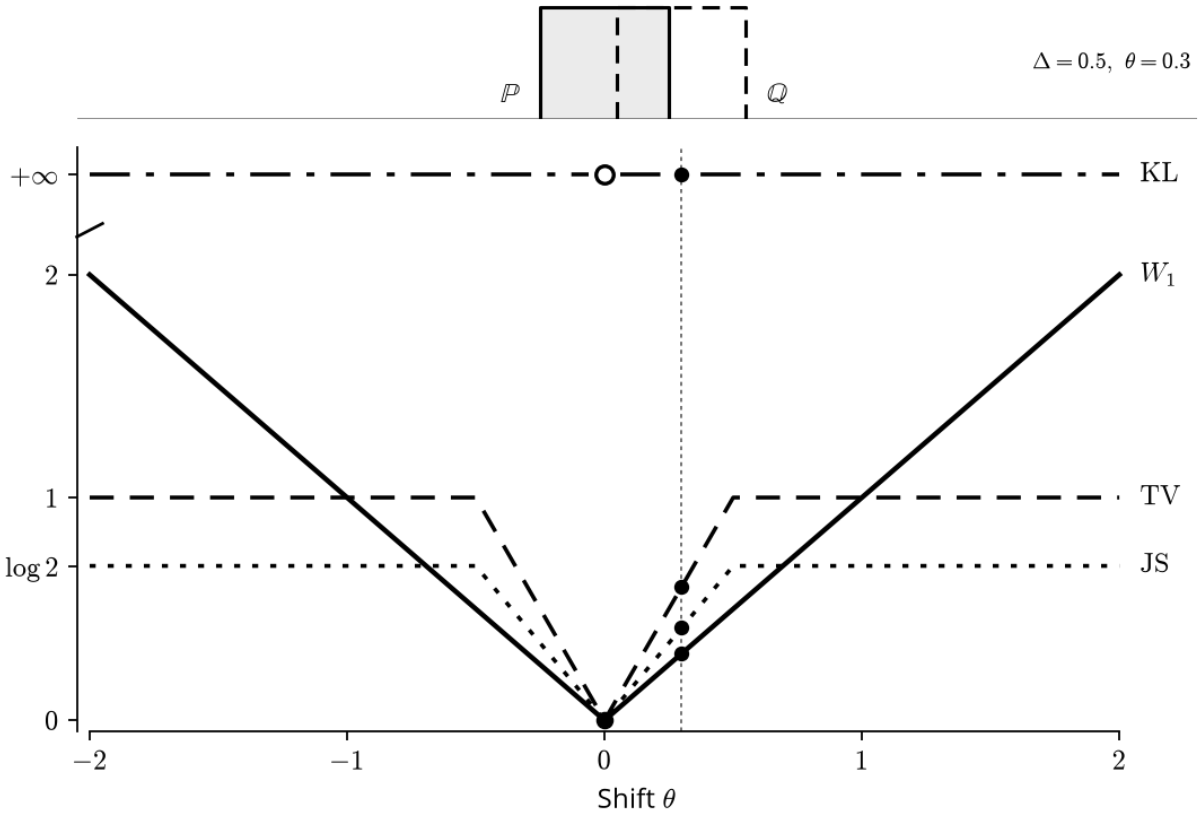



**Figure 1.** Two uniform distributions of width $\Delta$, $\mathbb{P}$ centred at $0$ and $\mathbb{Q}$ at $\theta$. While they overlap, TV and JS grow with the shift. Once they separate ($|\theta| \geq \Delta$) both stay at their maximum, and only $W_1$ keeps measuring the distance. KL is infinite as soon as $\theta \neq 0$. As $\Delta$ shrinks the uniforms approach point masses and the curves of TV and JS approach steps. The four quantities share one vertical axis, but $W_1$ is measured in units of $\theta$ while TV and JS have no unit, so their crossings carry no meaning. *The online version of this figure is interactive.*

Now suppose $\mathbb{Q}$ is the model, $\theta$ is its parameter and we want to move it onto $\mathbb{P}$ by gradient descent on one of these quantities. With KL the loss is infinite for every $\theta \neq 0$, even while the two uniforms still overlap, so there is no gradient to follow. With TV or JS the gradient points toward $\theta = 0$ while the supports overlap and vanishes once they separate. The distance $W_1$ gives a useful gradient for every $\theta \neq 0$, and so does $W_2$, as the box shows.

Going further

**The distance W2 on the two uniforms**

Besides $W_1 = |\theta|$ we also have $W_2 = |\theta|$, since $W_2 \geq W_1$ and the shift by $\theta$ moves every point by $|\theta|$. The inequality $W_2 \geq W_1$ holds because, under any coupling, $\mathbb{E}\|X - Y\|^2 \geq (\mathbb{E}\|X - Y\|)^2$.

A generator with $k < d$ is in the same situation as the two uniforms once they separate, because it usually shares no mass with the data. There are two cases, depending on what we assume about the data.

If the data have a density on $\mathbb{R}^d$, then by Chapter 1, Proposition 6 the image of the generator has volume zero, so the data give it probability $0$. If the data instead lie on a low-dimensional manifold, the image of the generator and the manifold are two thin sets. Two thin sets in

general position meet only in an even thinner set, as two lines in the plane cross at a single point (Arjovsky and Bottou 2017).

When the overlap carries no mass, $\mathrm{TV} = 1$, $\mathrm{JS} = \log 2$ and $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_g) = +\infty$. In the first case this holds for every setting of the weights. In the second it holds unless the image lines up with the manifold, as the image of the circle generator of Chapter 1 does, and then the two do share mass. While the two sets miss each other, moving the weights moves the image without changing TV or JS, so their gradient with respect to the weights is zero. This observation is the starting point of the Wasserstein GAN (Arjovsky, Chintala, and Bottou 2017), covered in Chapter 5.

Going further

**The support of a generator**

For a continuous generator whose latent distribution gives positive probability to every open set, as the Gaussian does, the support is the closure of the image $g(\mathbb{R}^k)$.

## Where the error of a trained model comes from

So far we have compared distributions as if we could compute these numbers exactly, and we cannot. We train on a stand-in computed from samples, such as the MMD to the training points in Chapter 1. Chapter 1 split the resulting error into four terms, and we would like to know how large they can be. This section bounds two of them together by a single inequality, the statistical term (we only have $n$ samples) and the surrogate term (the objective differs from the distance). The approximation term appears as it is, and we leave optimization aside.

As in Chapter 1, the error of a generator is $R(g) = D(\mathbb{P}^* \,\|\, \mathbb{P}_g)$ (the letter $R$ stands for *risk*, its usual name in learning theory). Fix also the objective $S_n(g)$ that we actually minimize, computed from the $n$ samples. Training returns $\hat{g} \in \arg\min_{g \in \mathcal{G}} S_n(g)$. We assume that a minimizer exists and that training finds it, so the bound below leaves out the optimization error. The idea of the bound is plain. If the objective is close to the true error for every generator, then minimizing the objective cannot go far wrong.

> **Proposition 5** (Error decomposition). *If $|S_n(g) - R(g)| \le \eta$ for every $g \in \mathcal{G}$, then*
>
> $$R(\hat{g}) \le \inf_{g \in \mathcal{G}} R(g) + 2\eta.$$

*Proof.*

For every $g \in \mathcal{G}$ we have $R(\hat{g}) \le S_n(\hat{g}) + \eta \le S_n(g) + \eta \le R(g) + 2\eta$. The first and last steps use the assumption and the middle step uses that $\hat{g}$ minimizes $S_n$. Take the infimum over $g$. □

In words, the trained generator is at most $2\eta$ worse than the best generator of the class. We pay $\eta$ once because the objective may flatter the generator we picked, and once more because it may be unfair to the best one.

The bound must hold for every $g$ at once because $\widehat{g}$ is chosen with the same samples that define $S_n$. A bound at a single fixed $g$ says nothing about the generator that training picks. The copying model shows the danger, since the objective of Chapter 1 is at its smallest there while its true error is not.

The deviation $\eta$ mixes two causes, and we can separate them. Write $S(g)$ for what $S_n(g)$ becomes with infinitely many samples. On the circle this is the difference between comparing with $12$ points and comparing with the whole circle. The triangle inequality $|S_n - R| \le |S_n - S| + |S - R|$ splits the deviation, and we get

$$R(\widehat{g}) \le \inf_{g \in \mathcal{G}} R(g) + 2 \sup_{g \in \mathcal{G}} |S_n(g) - S(g)| + 2 \sup_{g \in \mathcal{G}} |S(g) - R(g)|.$$

The first term is the approximation error of Chapter 1, what even the best generator of the class misses. We call the two suprema the *statistical deviation* and the *surrogate deviation*. The first comes from having finitely many samples, and the second from minimizing something other than the distance itself.

Going further

**How the deviations relate to the terms of Chapter 1**

With no optimization error, $\widehat{g}$ is the minimizer $g_{S_n}$ of Chapter 1, and the surrogate and statistical terms there add up to $R(\widehat{g}) - \inf_{g \in \mathcal{G}} R(g)$. The two deviations bound this sum together. The deviations are never negative, while the statistical term of Chapter 1 can be negative when its surrogate term is positive. The surrogate deviation is small only when $S$ is on the same scale as $R$. It can also be large when the surrogate error is zero. Adding a constant to $S$ leaves $\widehat{g}$ unchanged but adds that constant to $\sup_g |S(g) - R(g)|$. The negative log-likelihood of the last section differs from KL by such a constant, the entropy of $\mathbb{P}^*$, so the bound is applied after removing it.

The simplest case uses the distance itself as the objective, so the samples are the only stand-in. There $D$ is a metric $\rho$ and we minimize the distance to the samples themselves.

> **Corollary 6** (Fitting the empirical distribution). *If $\rho$ is a metric and $S_n(g) = \rho(\mathbb{P}_n, \mathbb{P}_g)$, then*
>
> $$\rho(\mathbb{P}^*, \mathbb{P}_{\widehat{g}}) \le \inf_{g \in \mathcal{G}} \rho(\mathbb{P}^*, \mathbb{P}_g) + 2\,\rho(\mathbb{P}_n, \mathbb{P}^*).$$

*Proof.*

By the triangle inequality $|\rho(\mathbb{P}_n, \mathbb{P}_g) - \rho(\mathbb{P}^*, \mathbb{P}_g)| \leq \rho(\mathbb{P}_n, \mathbb{P}^*)$ for every $g$. Apply Proposition 5 with $\eta = \rho(\mathbb{P}_n, \mathbb{P}^*)$. □

Beyond the approximation error, the guarantee now depends only on how close the samples are to the truth, a question we take up in the next section.

The approximation error and the two deviations are three different ways to miss $\mathbb{P}^*$, and they pull against each other. A larger model class $\mathcal{G}$ shrinks the approximation term but, with $n$ fixed, widens the statistical deviation. A surrogate that is cheap to estimate from samples may sit far from the true distance, widening the surrogate deviation. Every method in this series is one choice of where to spend this budget, through the distance $D$ it aims at and the surrogate $S_n$ it can minimize. The approximation term is a property of the class and will return with each architecture, and the statistical term is the gap that more training points narrowed in Chapter 1.

The surrogate deviation also hides something useful. The bound counts it in absolute value, as if $S$ could err on either side of $R$. For most surrogates we know something more precise. We know on which side of the distance they lie ($S \geq R$ or $S \leq R$), and the gap between them changes with $g$. The side matters. Take $n$ infinite for a moment and write $\mathrm{gap}(g) = |S(g) - R(g)|$.

> **Proposition 7** (Upper and lower surrogates). *Let $\hat{g}$ minimize $S$ over $\mathcal{G}$.*
>
> 1. If $S \geq R$ (an upper bound), then $R(\hat{g}) \leq R(g) + \mathrm{gap}(g)$ for every $g \in \mathcal{G}$.
> 2. If $S \leq R$ (a lower bound), then $R(\hat{g}) \leq R(g) + \mathrm{gap}(\hat{g})$ for every $g \in \mathcal{G}$.

*Proof.*

For an upper bound, $R(\hat{g}) \leq S(\hat{g}) \leq S(g) = R(g) + \mathrm{gap}(g)$. For a lower bound, $R(\hat{g}) = S(\hat{g}) + \mathrm{gap}(\hat{g}) \leq S(g) + \mathrm{gap}(\hat{g}) \leq R(g) + \mathrm{gap}(\hat{g})$. □

The two cases look alike on the page, and the difference is which generator carries the gap. An upper bound reads $S = R + \mathrm{gap}$, and lowering $S$ means lowering $R$ or the gap. The error is then controlled by the gap at any good model we care to compare with. A lower bound reads $S = R - \mathrm{gap}$. The optimizer can lower $S$ by raising the gap while $R$ stays where it is, and the error is controlled by the gap at the model that training selects. Minimizing a lower bound therefore favours the models whose distance to the data the surrogate underestimates the most.

Think of trying to lose weight with a faulty scale. A scale that only reads too high cannot make us think we are lighter than we are, since a low reading means a low weight. It may still hide the lowest weight we could reach. A scale that only reads too low rewards standing wherever it reads lowest, and the reading can fall while the weight stays the same.

Both kinds appear in this course. Variational autoencoders (Chapter 3) minimize an upper bound and GANs (Chapters 4 and 5) minimize lower bounds. This is the formal version of a familiar complaint about GANs, that the generator learns to exploit the blind spots of the discriminator.

Going further

**Which side each method lies on**

Variational autoencoders (Chapter 3) maximize the evidence lower bound, a lower bound on the log-likelihood, so they minimize an upper bound on the negative log-likelihood. Diffusion models trained with their variational weighting (Chapter 7) also minimize an upper bound. Flow matching (Chapter 8) minimizes a regression loss that controls $W_2$ up to a factor set by the smoothness of the learned model, which is weaker than a plain upper bound. GANs (Chapters 4 and 5) minimize lower bounds.

## What more samples buy us

We left one question open. In Corollary 6 the statistical deviation measures how far the empirical distribution is from the truth, and we would hope that more samples shrink it. For three of the four comparisons this term does not decrease with $n$.

> **Proposition 8** (Samples are always far in TV, JS and KL). *If $\mathbb{P}^*$ has no atoms (no single point has positive probability), then every $n$ gives $\mathrm{TV}(\mathbb{P}_n, \mathbb{P}^*) = 1$, $\mathrm{JS}(\mathbb{P}_n, \mathbb{P}^*) = \log 2$ and $\mathrm{KL}(\mathbb{P}_n \,\|\, \mathbb{P}^*) = +\infty$.*

*Proof.*

Take $A = \{X_1, \dots, X_n\}$. Then $\mathbb{P}_n(A) = 1$ and $\mathbb{P}^*(A) = 0$, so the two distributions share no mass. The supremum defining TV is therefore at least $1$, which is its maximum. For JS the mixture $\mathbb{M}$ gives each set half the mass of the distribution that lives on it. Each KL to $\mathbb{M}$ then equals $\log 2$, as for separated uniforms. Finally $A$ has positive mass under $\mathbb{P}_n$ and none under $\mathbb{P}^*$, so the KL is infinite. □

TV, JS and KL give the same value for ten samples and for a million, because they only see that $\mathbb{P}_n$ lives on finitely many points. The supports of $\mathbb{P}_n$ and $\mathbb{P}^*$ do meet here, since the samples come from $\mathbb{P}^*$ and lie in its support. The two distributions still share no mass, because $\mathbb{P}^*$ gives those points probability $0$. Disjoint supports are enough for TV to equal $1$, but they are not required.

The Wasserstein distance does shrink, and we can guess how fast. Spread $n$ points evenly over a surface of dimension $m$. Each point then has a patch holding about $1/n$ of the mass, and the side of that patch is about $n^{-1/m}$. To turn the samples into the true distribution, mass has to travel across a patch, so this distance falls no faster than $n^{-1/m}$.

Evenly spread points are the best case, and real samples are random. They crowd some regions and thin others, and evening that out costs about $n^{-1/2}$ in any dimension, the same size as the error of a sample mean. Above dimension $2$ the patch term is the larger one and sets the rate, and at dimension $2$ the two are about equal. On a curve the fluctuation term is the larger one, so a distribution on a curve ($m = 1$) gives $n^{-1/2}$. So the dimension of the data sets the rate, whatever the number of pixels. The manifold hypothesis of Chapter 1 therefore also decides how quickly more samples close the gap to the true distribution.

Going further

### The precise statement

Suppose $\mathbb{P}^*$ spreads its mass smoothly over a compact smooth manifold of dimension $m \geq 2$, in the sense of having a density with respect to the manifold's own surface measure (it has none on $\mathbb{R}^d$). Then $\mathbb{E}\, W_1(\mathbb{P}_n, \mathbb{P}^*)$ decays like $n^{-1/m}$ up to an arbitrarily small loss in the exponent, whatever the ambient dimension (Weed and Bach 2019, Theorem 1 with Propositions 7 and 9 of the arXiv version). Below dimension $2$ the rate stops improving at $n^{-1/2}$. Data that lie near a manifold follow the rate of the manifold's dimension until $n$ is large enough to resolve their spread around it. The next box gives the rates for any distribution on $\mathbb{R}^d$.

Going further

### Rates in d dimensions

For a distribution on $\mathbb{R}^d$ with a finite moment of some order larger than $2$, $\mathbb{E}\, W_1(\mathbb{P}_n, \mathbb{P}^*)$ is at most of order $n^{-1/2}$ when $d = 1$, $n^{-1/2} \log n$ when $d = 2$ and $n^{-1/d}$ when $d \geq 3$ (Fournier and Guillin 2015, Theorem 1). For the uniform distribution on a cube in dimension $d \geq 3$ it is no smaller than $n^{-1/d}$ (the third example after their Theorem 1). In low dimension the error comes from random fluctuations in how many points land in each region, which shrink like the error of a sample mean. In high dimension it comes from the distance mass must travel to fill the gaps between the samples.

These rates describe large $n$, so we also measure them on data. If the argument above is right, points on a curve should fall quickly, and a Gaussian with $16$ dimensions should fall slowly. The figure compares them with MNIST, and the box after it gives the method and the slopes.

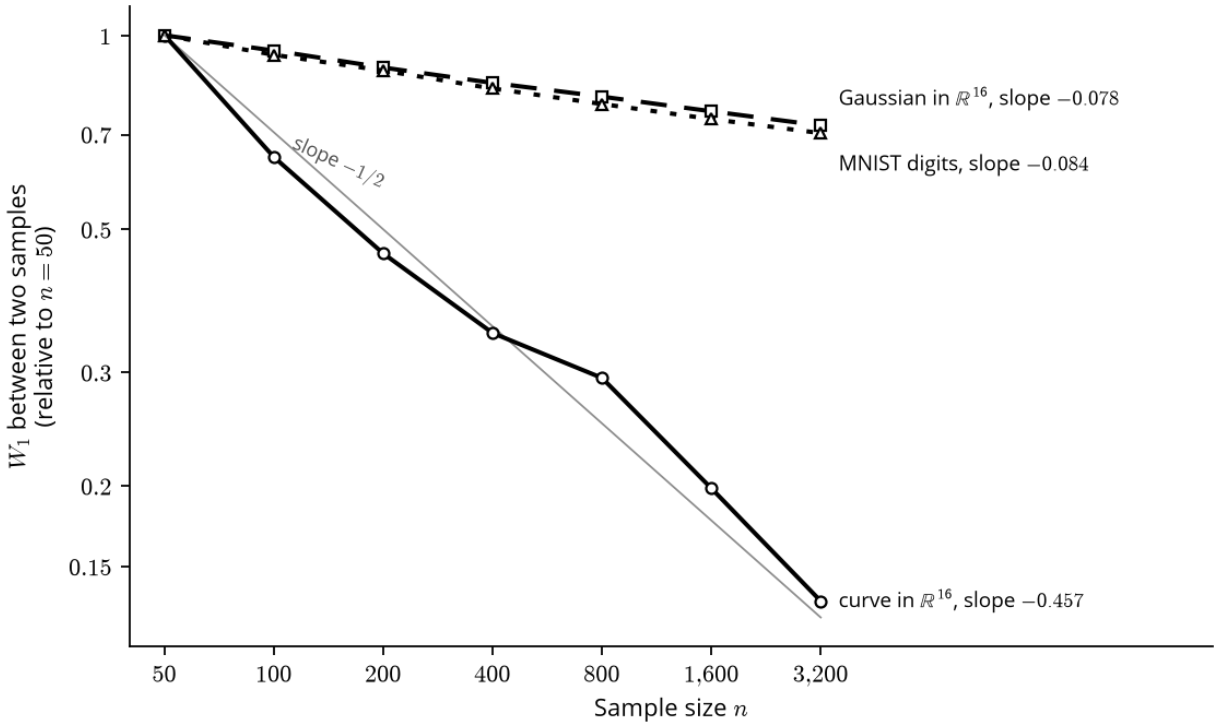



**Figure 2.** The Wasserstein distance between two independent samples of size $n$ from the same distribution, relative to its value at $n = 50$, on log-log axes. For points on a closed curve in $\mathbb{R}^{16}$ (intrinsic dimension 1) it falls like $n^{-1/2}$. For a Gaussian in $\mathbb{R}^{16}$ and for MNIST digits it falls by less than a third over a 64-fold increase of $n$.

Lab

### How quickly samples approach a distribution

We cannot compute $W_1(\mathbb{P}_n, \mathbb{P}^*)$ directly, because $\mathbb{P}^*$ is continuous and for MNIST unknown. We compute instead the $W_1$ between two samples of $n$ points each. For two sets of $n$ equally weighted points, $W_1$ is the average distance in the best one-to-one matching of the two sets. An assignment solver finds this matching exactly (we use `scipy.optimize.linear_sum_assignment`). The box after this one shows why this distance shrinks at the same rate as $W_1(\mathbb{P}_n, \mathbb{P}^*)$. We use $n$ from $50$ to $3{,}200$ with five repetitions each. For MNIST the two samples are disjoint random subsets of the $60{,}000$ training images. I added the Gaussian as a control, a case whose dimension is known, to see how far a dimension read off a slope can be trusted.

| Data | Intrinsic dimension | Slope | Expected |
|---|---|---|---|
| closed curve in $\mathbb{R}^{16}$ | 1 | $-0.457$ | $-1/2$, the rate of dimension 1 |
| Gaussian in $\mathbb{R}^{16}$ | 16 | $-0.078$ | $-1/16 \approx -0.063$ (upper-bound rate) |
| MNIST digits ($d = 784$) | unknown | $-0.084$ | |

I expected MNIST to fall clearly faster than the Gaussian, since its intrinsic dimension is near $10$ and the Gaussian's is $16$. The two slopes came out close.

The curve and the Gaussian live in the same $\mathbb{R}^{16}$ and differ only in their intrinsic dimension, which changes the rate. Reading a slope $-1/m$ as a dimension $m$ gives $13$ for the Gaussian, whose true dimension is $16$. For MNIST it gives about $12$, inside the range of $7$ to $13$ that Pope et al. (2021) measure with a different estimator. The Gaussian shows that at these sample sizes the reading is low, so the true value for MNIST may be higher. Read these dimensions only as rough magnitudes.

Going further

**Why the rate between two samples is the rate to the truth**

Let $\mathbb{P}_n$ and $\mathbb{P}'_n$ be the empirical distributions of two independent samples of $\mathbb{P}^*$. The triangle inequality gives $W_1(\mathbb{P}_n, \mathbb{P}'_n) \le W_1(\mathbb{P}_n, \mathbb{P}^*) + W_1(\mathbb{P}^*, \mathbb{P}'_n)$, so the expected distance between the two samples is at most twice $\mathbb{E}\, W_1(\mathbb{P}_n, \mathbb{P}^*)$.

For the other direction, fix the first sample. For each draw of the second sample, take an optimal transport plan from $\mathbb{P}_n$ to $\mathbb{P}'_n$. Average these plans over all draws of the second sample. The average is a transport plan from $\mathbb{P}_n$ to $\mathbb{P}^*$, because $\mathbb{E}[\mathbb{P}'_n(A)] = \mathbb{P}^*(A)$ for every set $A$, and its cost is the average of the costs. So $W_1(\mathbb{P}_n, \mathbb{P}^*)$ is at most the expected distance between the two samples. The two expectations are within a factor $2$ of each other and have the same rate.

Between two sets of $n$ equally weighted points, a transport plan is an $n \times n$ matrix with nonnegative entries whose rows and columns sum to $1/n$. Birkhoff's theorem says that every such matrix is an average of permutation matrices, scaled by $1/n$. The cost is linear in the plan, so its minimum is reached at a permutation, which is a one-to-one matching. For MNIST the samples are drawn without replacement from the training set, and the same argument applies with the training set in place of $\mathbb{P}^*$.

These slopes are small, and it helps to see what that means. At a rate of roughly $n^{-1/12}$, halving the distance between the samples and $\mathbb{P}^*$ takes $2^{12} \approx 4{,}000$ times more data. CIFAR-10 has an intrinsic dimension of $13$ to $26$ (Chapter 1) and needs even more. This rate is the statistical deviation $2\, W_1(\mathbb{P}_n, \mathbb{P}^*)$ of Corollary 6, so fitting the samples in $W_1$ guarantees little with realistic $n$. Over a class that contains $\mathbb{P}_n$, the best fit to the samples is the copying model itself. What a model does between the samples therefore has to come from the model class and the training objective, as the circle of Chapter 1 showed.

Going further

**Why not train on the Wasserstein distance directly**

A generator could be trained on $W_1$ directly, by matching a batch of generated samples to a batch of data with an assignment solver. This is expensive. Each step costs an exact matching, which grows roughly like the cube of the batch size, and the estimate from a batch inherits the slow rate measured above. Chapter 5 estimates $W_1$ through a second network. Another option compares distributions only through a restricted family of test functions, such as kernels or

neural networks. The MMD that trained the circle generators of Chapter 1 is one of them. Such distances can be estimated quickly from samples, but some of them cannot tell a copy of the training set from the truth, as the next box shows.

Going further

**Restricted distances with fast rates and blind spots**

Compare distributions only through a family $\mathcal{F}$ of test functions,

$$\rho_{\mathcal{F}}(\mathbb{P}, \mathbb{Q}) = \sup_{u \in \mathcal{F}} \left| \mathbb{E}_{\mathbb{P}} u - \mathbb{E}_{\mathbb{Q}} u \right|.$$

This is symmetric and satisfies the triangle inequality. It is a metric only when $\mathcal{F}$ is rich enough to separate any two different distributions. Otherwise two different distributions can be at distance $0$.

Take for $\mathcal{F}$ a class of neural networks with $N$ parameters, with bounded outputs and Lipschitz in their parameters. Arora et al. (2017) show that with probability at least $1 - \exp(-N)$, the empirical estimate of $\rho_{\mathcal{F}}$ is within $\varepsilon$ of its true value once the number of samples is of order $N \log(N/\varepsilon)/\varepsilon^2$ (their Theorem 3.1). The rate depends on the size of the networks and does not depend on the dimension of the data. Their next corollary shows the cost (Corollary 3.1 of the conference version, 3.2 of the arXiv version). With that many samples the empirical distribution $\mathbb{P}_n$ itself is within $\varepsilon$ of $\mathbb{P}^*$, so a restricted distance of this kind cannot tell a model that copies its training set from the truth. The Wasserstein GAN of Chapter 5 measures a distance of this form, and the original GAN of Chapter 4 a variant with a logarithm.

The maximum mean discrepancy (MMD) used in Chapter 1's lab takes for $\mathcal{F}$ the unit ball of the function space attached to a kernel $\kappa$, here a sum of Gaussian kernels. For independent $X, X' \sim \mathbb{P}$ and $Y, Y' \sim \mathbb{Q}$ it has the closed form

$$\rho_{\mathcal{F}}(\mathbb{P}, \mathbb{Q})^2 = \mathbb{E}\,\kappa(X, X') + \mathbb{E}\,\kappa(Y, Y') - 2\,\mathbb{E}\,\kappa(X, Y),$$

which samples estimate directly. With Gaussian kernels this family separates distributions, and MMD is a metric.

## Maximum likelihood is a KL divergence

KL has now failed twice. It is infinite between a generator and the data, and between the samples and the truth. Yet the most common way to fit a model $\mathbb{P}_\theta$ with a density $p_\theta$ minimizes a KL divergence, and this section explains how the two facts fit together. The method is maximum likelihood, which maximizes the average log-likelihood $\frac{1}{n}\sum_{i=1}^{n} \log p_\theta(X_i)$ of the training images. In this section a density is a density on $\mathbb{R}^d$ in the sense of Chapter 1, which is the case $\nu$ = volume of the previous sections.

**Proposition 9** (Maximum likelihood minimizes KL). *Let $\mathbb{P}^*$ have density $p^*$, and assume $\mathbb{E}_{X\sim\mathbb{P}^*}|\log p^*(X)|$ and $\mathbb{E}_{X\sim\mathbb{P}^*}|\log p_\theta(X)|$ are finite. Then*

$$\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta) = \mathbb{E}_{X\sim\mathbb{P}^*}[\log p^*(X)] - \mathbb{E}_{X\sim\mathbb{P}^*}[\log p_\theta(X)],$$

*so minimizing $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ over $\theta$ is the same as maximizing $\mathbb{E}_{X\sim\mathbb{P}^*}[\log p_\theta(X)]$.*

*Proof.*

Split the logarithm of the ratio in the definition of KL. The first term does not depend on $\theta$. □

The first term in the identity describes the data and is the same for every model, so only the second term can move. The average log-likelihood over the training images is a sample mean that estimates $\mathbb{E}_{X\sim\mathbb{P}^*}[\log p_\theta(X)]$, so maximum likelihood minimizes an estimate of $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ up to the constant $\mathbb{E}\log p^*(X)$. Adding to the objective a constant that does not depend on the model does not change the trained model or its error, so the negative log-likelihood can stand for $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ in Proposition 5 once that constant is removed.

Now we can see why the second failure of KL does not make the objective of maximum likelihood infinite. Plugging $\mathbb{P}_n$ in place of $\mathbb{P}^*$ would fail here too, since $\mathrm{KL}(\mathbb{P}_n \,\|\, \mathbb{P}_\theta) = +\infty$ for the reason of Proposition 8. A model with a density gives the training points probability $0$. The order of the steps saves us. Maximum likelihood first writes the KL with the density $p_\theta$, as in Proposition 9, and only then replaces the expectation over $\mathbb{P}^*$ by an average over the samples. With $\mathbb{P}_n$ in place of $\mathbb{P}^*$, the part of the split that breaks is the first term, because $\mathbb{P}_n$ has no density on $\mathbb{R}^d$. That term does not involve $\theta$ and has already been set aside.

Going further

**What the sample average promises**

For each fixed $\theta$ with $\mathbb{E}|\log p_\theta(X)|$ finite, the average converges like any sample mean. This does not control the model that training picks, and a class that can put narrow spikes on the training points has an unbounded likelihood.

The first failure does affect maximum likelihood. A generator with $k < d$ has no density, and Chapter 3 adds noise to the generator to obtain one.

Maximum likelihood also fixes the direction of the KL, with the data first, and this direction affects what the model learns. In $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ the expectation runs over the data. A data point to which the model gives almost no density costs a lot, since $\log p_\theta(X) \to -\infty$. A region where the model puts mass and no data lives costs nothing directly, since the expectation never visits it. It costs only indirectly, through the density it takes away from the data. The model is therefore pushed to *cover* every data point, even if it has to put mass between them.

The reverse direction $\mathrm{KL}(\mathbb{P}_\theta \,\|\, \mathbb{P}^*)$ flips these costs. It averages $\log(p_\theta/p^*)$ over samples of the model. A model sample where there are no data costs $+\infty$, and an ignored region of the data costs nothing directly. A model trained in this direction tends to concentrate on one *mode*, a single peak of the data distribution.

We can see the difference most clearly when the model is too simple to fit the data, as when a single Gaussian is fitted to a target with two modes. In the direction of maximum likelihood the answer has a closed form.

> **Proposition 10** (Forward KL fits moments). *Let $\mathbb{P}$ have a density $p$ with $\mathbb{E}_{\mathbb{P}}|\log p(X)| < \infty$ and a finite second moment, and write $\mu_{\mathbb{P}}$ and $\Sigma_{\mathbb{P}}$ for its mean and covariance ($\Sigma_{\mathbb{P}}$ is invertible because $\mathbb{P}$ has a density, and a distribution confined to a flat subspace has none). Among Gaussians $\mathcal{N}(\mu, \Sigma)$, the unique minimizer of $\mathrm{KL}(\mathbb{P} \,\|\, \mathcal{N}(\mu, \Sigma))$ is $\mathcal{N}(\mu_{\mathbb{P}}, \Sigma_{\mathbb{P}})$.*

*Proof.*

Under these assumptions $\mathrm{KL}(\mathbb{P} \,\|\, \mathcal{N}(\mu, \Sigma))$ equals a finite constant plus $-\mathbb{E}_{\mathbb{P}}[\log \mathcal{N}(X; \mu, \Sigma)] = \frac{1}{2}\log\det(2\pi\Sigma) + \frac{1}{2}\mathbb{E}_{\mathbb{P}}[(X-\mu)^\top \Sigma^{-1}(X-\mu)]$. The expectation equals $\mathrm{tr}(\Sigma^{-1}\Sigma_{\mathbb{P}}) + (\mu - \mu_{\mathbb{P}})^\top \Sigma^{-1}(\mu - \mu_{\mathbb{P}})$, which is minimized in $\mu$ only at $\mu = \mu_{\mathbb{P}}$. Write $\Lambda = \Sigma^{-1}$. What remains is $\frac{1}{2}(-\log\det\Lambda + \mathrm{tr}(\Lambda\Sigma_{\mathbb{P}}))$ up to a constant. This is a strictly convex function of $\Lambda$, and its gradient $\frac{1}{2}(-\Lambda^{-1} + \Sigma_{\mathbb{P}})$ vanishes only at $\Lambda = \Sigma_{\mathbb{P}}^{-1}$. □

So the best Gaussian in this direction only has to match the mean and the covariance of the target, whatever shape the target has. On a target with two modes, that gives a wide Gaussian centred between them. The reverse direction has no such formula, and the figure and the box below show what each one does.

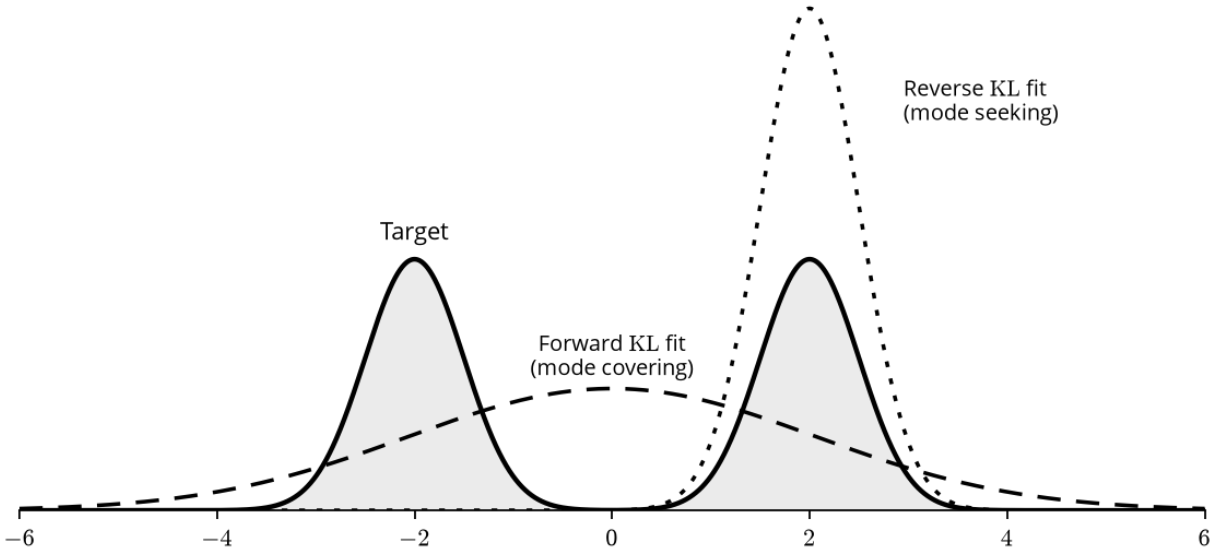



**Figure 3.** One Gaussian fitted to a target with two modes, $\frac{1}{2}\mathcal{N}(-2, 0.5^2) + \frac{1}{2}\mathcal{N}(2, 0.5^2)$, drawn as the solid curve. The direction of maximum likelihood, minimizing $\mathrm{KL}(\mathrm{target} \,\|\, q)$, gives the wide dashed Gaussian. It covers both modes and puts mass between them. Minimizing $\mathrm{KL}(q \,\|\, \mathrm{target})$ gives the dotted Gaussian, which settles on the mode at $+2$. It is twice as tall as that mode because it carries all the mass the target splits in two.

Lab

### The two directions of KL

The forward fit is the moment-matching Gaussian of Proposition 10. Its mean is $0$, and its variance $4.25$ adds the variance $0.5^2$ within each mode to the variance $2^2$ between the modes, so its standard deviation is $2.06$. A direct numerical minimization finds the same values. The reverse fit is minimized numerically from a start at $\mu = 1$ and lands on the mode at $+2$ with $\sigma = 0.50$ ( $0.5004$ before rounding, which the forward KL below reflects). I started at $\mu = 1$ because at $\mu = 0$ the gradient in $\mu$ is zero by symmetry, so the fit has no reason to pick a mode. By the same symmetry, a start at $\mu = -1$ lands on the mode at $-2$. Each fit has a large divergence in the other direction.

| Fit | $\mathrm{KL}(\mathrm{target} \,\|\, q)$ | $\mathrm{KL}(q \,\|\, \mathrm{target})$ |
|---|---|---|
| forward (covers both modes) | 0.72 | 2.10 |
| reverse (one mode) | 15.28 | 0.69 |

The reverse fit puts all its mass where the target puts half, so on its support the ratio $q/\mathrm{target}$ is about $2$ and its reverse KL is $\log 2$. The forward fit pays for the mass it puts between the modes.

The two behaviours are called *mode covering* and *mode seeking*. A model trained by maximum likelihood that cannot fit every mode spreads mass between them. Chapter 3 shows this on a VAE trained on eight separated clusters, where some samples land between them. The blur of VAE images has another source, which that chapter also explains.

Going further

### Covering and seeking in later models

Autoregressive models trained by exact maximum likelihood produce sharp samples (Oord, Kalchbrenner, and Kavukcuoglu 2016), so maximum likelihood alone does not make samples blurry (Theis, Oord, and Bethge 2016). GANs tend to drop modes and produce sharp samples although they minimize neither direction of KL (Chapter 4). Chapter 6 measures the two failures separately.

In practice

Each method of this series works with one of the distances above. Variational autoencoders minimize an upper bound on $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ up to a constant (Chapter 3), and diffusion models a weighted version of such a bound (Chapter 7). The original GAN minimizes a lower bound of $2\,\mathrm{JS} - \log 4$ (Chapter 4) and the Wasserstein GAN a lower bound of $W_1$ (Chapter 5). The FID score is the squared $W_2$ between Gaussians fitted to features of an image classifier, and precision and recall measure the two failures of the KL directions separately (Chapter 6). Flow matching minimizes a regression loss that controls $W_2$ (Chapter 8).

Check yourself

1. What is the total variation between two independent samples of the same distribution without atoms?

   Answer

   > It is $1$. Almost surely no point of one sample is a point of the other, so the set of points of the first sample has probability $1$ under one empirical distribution and $0$ under the other.

2. A training objective is a lower bound of the distance we care about. Why is minimizing it risky?

   Answer

   > Write it as $S = R - \mathrm{gap}$. The optimizer can lower $S$ by raising the gap, so by the proposition on upper and lower surrogates the error of the trained model is controlled by the gap at the model that training selects.

3. Maximum likelihood corresponds to $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$. Why that direction?

   Answer

   > The log-likelihood is an average over the data. By Proposition 9, maximizing it minimizes the KL whose expectation runs over $\mathbb{P}^*$. The reverse KL averages over the model and would require the data density.

## Summary

We wanted a number that says how far a generator is from the data, and the plain candidates disappointed us. TV, JS and KL compare densities point by point, so they cannot see how far apart the unshared mass lies. On two shifted uniforms, KL is infinite for any nonzero shift and TV and JS stop changing once the supports separate, while the Wasserstein distance keeps measuring the shift. A generator with fewer latent dimensions than pixels shares no mass with the data, so of the four comparisons only Wasserstein gives it a useful gradient.

We then asked what training on samples costs. The error of a trained model is at most the approximation error plus twice the statistical deviation plus twice the surrogate deviation, and minimizing an upper bound on the distance is safer than minimizing a lower bound. More samples do not help in TV, JS or KL, where the empirical distribution never gets closer to $\mathbb{P}^*$. In Wasserstein they help slowly, at a rate close to $n^{-1/m}$ for a distribution spread over an $m$-dimensional manifold with $m \geq 2$, and on MNIST we measured a slope near $-1/12$ for $n$ up to $3{,}200$.

KL returned in the last section, because maximum likelihood minimizes an estimate of $\mathrm{KL}(\mathbb{P}^* \,\|\, \mathbb{P}_\theta)$ without ever plugging $\mathbb{P}_n$ into a KL. This direction pushes the model to cover every mode, while the reverse direction tends to concentrate on one.

One obstacle remains. Maximum likelihood needs a density, and a generator with fewer latent dimensions than pixels has none. Chapter 3 gives the generator a density by adding a little noise and trains it by maximizing a lower bound on the likelihood, and it shows why the samples of

that model come out blurry.

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